

Human Review: Optimal Bourgain constant

Original automatic proof: `solution_packet.pdf`

Checked date: 14 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proposed proof appears to give a correct improvement of the constant in Rodriguez's Theorem 2.11 from 2 to 1. The key point is identifying exactly where the factor ε is lost in the original proof and replacing that estimate by a sharper midpoint estimate.

In Rodriguez's Lemma 2.17, one considers scalar functions of the form

$$a_k(t) = \langle z_{n_k}, \varphi(t) \rangle$$

whose pointwise oscillation is bounded by ε :

$$\limsup_k a_k(t) - \liminf_k a_k(t) \leq \varepsilon \quad \text{for a.e. } t.$$

The original proof centers $a_k(t)$ at the endpoint

$$\liminf_k a_k(t).$$

This controls the error by the full oscillation, and hence gives an ε -bound.

The corrected argument centers instead at the midpoint

$$m(t) = \frac{1}{2} \left(\limsup_k a_k(t) + \liminf_k a_k(t) \right).$$

Then

$$\limsup_k |a_k(t) - m(t)| \leq \frac{1}{2} \left(\limsup_k a_k(t) - \liminf_k a_k(t) \right) \leq \frac{\varepsilon}{2}$$

for almost every t . Applying Rodriguez's reverse Fatou lemma, Lemma 2.13 of the source paper, gives

$$\limsup_k \int_0^1 |a_k(t) - m(t)| dt \leq \frac{\varepsilon}{2}.$$

Improved Lemma 2.17

With this midpoint replacement, the proof of Rodriguez's Lemma 2.17 gives

$$\text{clust}_{L_1([0,1],Z)^{**}}(r_n \otimes z_n) \subseteq \frac{\varepsilon}{2} B_{L_1([0,1],Z)^{**}}$$

instead of the original εB -bound.

The last step in the estimate is unchanged. One writes

$$a_k(t) = m(t) + (a_k(t) - m(t)).$$

Then

$$|\langle r_{n_k} \otimes z_{n_k}, \varphi \rangle| \leq \left| \int_0^1 r_{n_k}(t) m(t) dt \right| + \int_0^1 |a_k(t) - m(t)| dt.$$

The first term tends to zero because the Rademacher sequence is weakly null in $L_1[0,1]$ and $m \in L_\infty[0,1]$. The second term is bounded by $\varepsilon/2$ in limsup by the midpoint-Fatou estimate above.

Propagation Through the Original Proof

Once this sharpened version of Lemma 2.17 is available, the rest of the proof is essentially Rodriguez's original proof of Theorem 2.11. The pointwise cluster estimate is passed to the Bochner L_1 -space using Rodriguez's Lemma 2.18. Then Lemma 2.16 excludes uniformly integrable ℓ_1 -sequences with constant $C > \varepsilon/2$. The general uniformly integrable case is handled by the same truncation argument as in the source paper.

Finally, the endpoint passage from all $\varepsilon' > \varepsilon$ back to ε uses Rodriguez's existing closedness lemma for ε -weak precompactness. The one-point probability space example with the scaled unit vector basis of ℓ_1 shows that the constant 1 cannot be improved.

Review Outcome

Archive this packet as an apparently correct full solution to Question 2.21: the constant 2 is not optimal, and the optimal universal constant is 1. The mathematical content is concentrated in the midpoint correction to the scalar oscillation estimate; after that correction, the argument follows the architecture of the original paper.